
Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

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Abstract

Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks. However, their ability to access and precisely manipulate knowledge is still limited, and hence on knowledge-intensive tasks, their performance lags behind task-specific architectures. Additionally, providing provenance for their decisions and updating their world knowledge remain open research problems. Pre-trained models with a differentiable access mechanism to explicit non-parametric memory have so far been only investigated for extractive downstream tasks. We explore a general-purpose fine-tuning recipe for retrieval-augmented generation (RAG) — models which combine pre-trained parametric and non-parametric memory for language generation. We introduce RAG models where the parametric memory is a pre-trained seq2seq model and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever. We compare two RAG formulations, one which conditions on the same retrieved passages across the whole generated sequence, and another which can use different passages per token. We fine-tune and evaluate our models on a wide range of knowledge-intensive NLP tasks and set the state of the art on three open domain QA tasks, outperforming parametric seq2seq models and task-specific retrieve-and-extract architectures. For language generation tasks, we find that RAG models generate more specific, diverse and factual language than a state-of-the-art parametric-only seq2seq baseline.

1 Introduction

Pre-trained neural language models have been shown to learn a substantial amount of in-depth knowledge from data [47]. They can do so without any access to an external memory, as a parameterized implicit knowledge base [51, 52]. While this development is exciting, such models do have downsides: They cannot easily expand or revise their memory, can’t straightforwardly provide insight into their predictions, and may produce “hallucinations” [38]. Hybrid models that combine parametric memory with non-parametric (i.e., retrieval-based) memories [20, 26, 48] can address some of these issues because knowledge can be directly revised and expanded, and accessed knowledge can be inspected and interpreted. REALM [20] and ORQA [31], two recently introduced models that combine masked language models [8] with a differentiable retriever, have shown promising results,

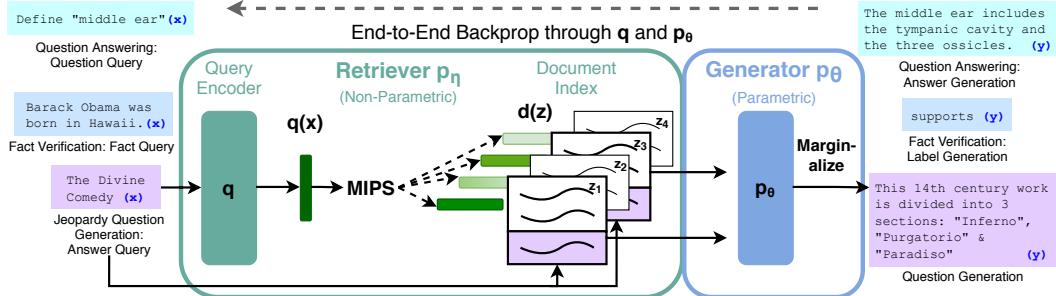


Figure 1: Overview of our approach. We combine a pre-trained retriever (*Query Encoder* + *Document Index*) with a pre-trained seq2seq model (*Generator*) and fine-tune end-to-end. For query x , we use Maximum Inner Product Search (MIPS) to find the top-K documents z_i . For final prediction y , we treat z as a latent variable and marginalize over seq2seq predictions given different documents.

but have only explored open-domain extractive question answering. Here, we bring hybrid parametric and non-parametric memory to the “workhorse of NLP,” i.e. sequence-to-sequence (seq2seq) models.

We endow pre-trained, parametric-memory generation models with a non-parametric memory through a general-purpose fine-tuning approach which we refer to as retrieval-augmented generation (RAG). We build RAG models where the parametric memory is a pre-trained seq2seq transformer, and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever. We combine these components in a probabilistic model trained end-to-end (Fig. 1). The retriever (Dense Passage Retriever [26], henceforth DPR) provides latent documents conditioned on the input, and the seq2seq model (BART [32]) then conditions on these latent documents together with the input to generate the output. We marginalize the latent documents with a top-K approximation, either on a per-output basis (assuming the same document is responsible for all tokens) or a per-token basis (where different documents are responsible for different tokens). Like T5 [51] or BART, RAG can be fine-tuned on any seq2seq task, whereby both the generator and retriever are jointly learned.

There has been extensive previous work proposing architectures to enrich systems with non-parametric memory which are trained from scratch for specific tasks, e.g. memory networks [64, 55], stack-augmented networks [25] and memory layers [30]. In contrast, we explore a setting where both parametric and non-parametric memory components are pre-trained and pre-loaded with extensive knowledge. Crucially, by using pre-trained access mechanisms, the ability to access knowledge is present without additional training.

Our results highlight the benefits of combining parametric and non-parametric memory with generation for *knowledge-intensive tasks*—tasks that humans could not reasonably be expected to perform without access to an external knowledge source. Our RAG models achieve state-of-the-art results on open Natural Questions [29], WebQuestions [3] and CuratedTrec [2] and strongly outperform recent approaches that use specialised pre-training objectives on TriviaQA [24]. Despite these being extractive tasks, we find that unconstrained generation outperforms previous extractive approaches. For knowledge-intensive generation, we experiment with MS-MARCO [1] and Jeopardy question generation, and we find that our models generate responses that are more factual, specific, and diverse than a BART baseline. For FEVER [56] fact verification, we achieve results within 4.3% of state-of-the-art pipeline models which use strong retrieval supervision. Finally, we demonstrate that the non-parametric memory can be replaced to update the models’ knowledge as the world changes.¹

2 Methods

We explore RAG models, which use the input sequence x to retrieve text documents z and use them as additional context when generating the target sequence y . As shown in Figure 1, our models leverage two components: (i) a retriever $p_\eta(z|x)$ with parameters η that returns (top-K truncated) distributions over text passages given a query x and (ii) a generator $p_\theta(y_i|x, z, y_{1:i-1})$ parametrized

¹Code to run experiments with RAG has been open-sourced as part of the HuggingFace Transformers Library [66] and can be found at <https://github.com/huggingface/transformers/blob/master/examples/rag/>. An interactive demo of RAG models can be found at <https://huggingface.co/rag/>

by θ that generates a current token based on a context of the previous $i - 1$ tokens $y_{1:i-1}$, the original input x and a retrieved passage z .

To train the retriever and generator end-to-end, we treat the retrieved document as a latent variable. We propose two models that marginalize over the latent documents in different ways to produce a distribution over generated text. In one approach, *RAG-Sequence*, the model uses the same document to predict each target token. The second approach, *RAG-Token*, can predict each target token based on a different document. In the following, we formally introduce both models and then describe the p_η and p_θ components, as well as the training and decoding procedure.

2.1 Models

RAG-Sequence Model The RAG-Sequence model uses the same retrieved document to generate the complete *sequence*. Technically, it treats the retrieved document as a single latent variable that is marginalized to get the seq2seq probability $p(y|x)$ via a top-K approximation. Concretely, the top K documents are retrieved using the retriever, and the generator produces the output sequence probability for each document, which are then marginalized,

$$p_{\text{RAG-Sequence}}(y|x) \approx \sum_{z \in \text{top-}k(p(\cdot|x))} p_\eta(z|x) p_\theta(y|x, z) = \sum_{z \in \text{top-}k(p(\cdot|x))} p_\eta(z|x) \prod_i^N p_\theta(y_i|x, z, y_{1:i-1})$$

RAG-Token Model In the RAG-Token model we can draw a different latent document for each target *token* and marginalize accordingly. This allows the generator to choose content from several documents when producing an answer. Concretely, the top K documents are retrieved using the retriever, and then the generator produces a distribution for the next output token for each document, before marginalizing, and repeating the process with the following output token. Formally, we define:

$$p_{\text{RAG-Token}}(y|x) \approx \prod_i^N \sum_{z \in \text{top-}k(p(\cdot|x))} p_\eta(z|x) p_\theta(y_i|x, z, y_{1:i-1})$$

Finally, we note that RAG can be used for sequence classification tasks by considering the target class as a target sequence of length one, in which case RAG-Sequence and RAG-Token are equivalent.

2.2 Retriever: DPR

The retrieval component $p_\eta(z|x)$ is based on DPR [26]. DPR follows a bi-encoder architecture:

$$p_\eta(z|x) \propto \exp(\mathbf{d}(z)^\top \mathbf{q}(x)) \quad \mathbf{d}(z) = \text{BERT}_d(z), \quad \mathbf{q}(x) = \text{BERT}_q(x)$$

where $\mathbf{d}(z)$ is a dense representation of a document produced by a $\text{BERT}_{\text{BASE}}$ *document encoder* [8], and $\mathbf{q}(x)$ a query representation produced by a *query encoder*, also based on $\text{BERT}_{\text{BASE}}$. Calculating $\text{top-}k(p_\eta(\cdot|x))$, the list of k documents z with highest prior probability $p_\eta(z|x)$, is a Maximum Inner Product Search (MIPS) problem, which can be approximately solved in sub-linear time [23]. We use a pre-trained bi-encoder from DPR to initialize our retriever and to build the document index. This retriever was trained to retrieve documents which contain answers to TriviaQA [24] questions and Natural Questions [29]. We refer to the document index as the *non-parametric memory*.

2.3 Generator: BART

The generator component $p_\theta(y_i|x, z, y_{1:i-1})$ could be modelled using any encoder-decoder. We use BART-large [32], a pre-trained seq2seq transformer [58] with 400M parameters. To combine the input x with the retrieved content z when generating from BART, we simply concatenate them. BART was pre-trained using a denoising objective and a variety of different noising functions. It has obtained state-of-the-art results on a diverse set of generation tasks and outperforms comparably-sized T5 models [32]. We refer to the BART generator parameters θ as the *parametric memory* henceforth.

2.4 Training

We jointly train the retriever and generator components without any direct supervision on what document should be retrieved. Given a fine-tuning training corpus of input/output pairs (x_j, y_j) , we

minimize the negative marginal log-likelihood of each target, $\sum_j -\log p(y_j|x_j)$ using stochastic gradient descent with Adam [28]. Updating the document encoder BERT_d during training is costly as it requires the document index to be periodically updated as REALM does during pre-training [20]. We do not find this step necessary for strong performance, and keep the document encoder (and index) fixed, only fine-tuning the query encoder BERT_q and the BART generator.

2.5 Decoding

At test time, RAG-Sequence and RAG-Token require different ways to approximate $\arg \max_y p(y|x)$.

RAG-Token The RAG-Token model can be seen as a standard, autoregressive seq2seq generator with transition probability: $p'_\theta(y_i|x, y_{1:i-1}) = \sum_{z \in \text{top-}k(p(\cdot|x))} p_\eta(z_i|x) p_\theta(y_i|x, z_i, y_{1:i-1})$. To decode, we can plug $p'_\theta(y_i|x, y_{1:i-1})$ into a standard beam decoder.

RAG-Sequence For RAG-Sequence, the likelihood $p(y|x)$ does not break into a conventional per-token likelihood, hence we cannot solve it with a single beam search. Instead, we run beam search for each document z , scoring each hypothesis using $p_\theta(y_i|x, z, y_{1:i-1})$. This yields a set of hypotheses Y , some of which may not have appeared in the beams of all documents. To estimate the probability of an hypothesis y we run an additional forward pass for each document z for which y does not appear in the beam, multiply generator probability with $p_\eta(z|x)$ and then sum the probabilities across beams for the marginals. We refer to this decoding procedure as “Thorough Decoding.” For longer output sequences, $|Y|$ can become large, requiring many forward passes. For more efficient decoding, we can make a further approximation that $p_\theta(y|x, z_i) \approx 0$ where y was not generated during beam search from x, z_i . This avoids the need to run additional forward passes once the candidate set Y has been generated. We refer to this decoding procedure as “Fast Decoding.”

3 Experiments

We experiment with RAG in a wide range of knowledge-intensive tasks. For all experiments, we use a single Wikipedia dump for our non-parametric knowledge source. Following Lee et al. [31] and Karpukhin et al. [26], we use the December 2018 dump. Each Wikipedia article is split into disjoint 100-word chunks, to make a total of 21M documents. We use the document encoder to compute an embedding for each document, and build a single MIPS index using FAISS [23] with a Hierarchical Navigable Small World approximation for fast retrieval [37]. During training, we retrieve the top k documents for each query. We consider $k \in \{5, 10\}$ for training and set k for test time using dev data. We now discuss experimental details for each task.

3.1 Open-domain Question Answering

Open-domain question answering (QA) is an important real-world application and common testbed for knowledge-intensive tasks [20]. We treat questions and answers as input-output text pairs (x, y) and train RAG by directly minimizing the negative log-likelihood of answers. We compare RAG to the popular extractive QA paradigm [5, 7, 31, 26], where answers are extracted spans from retrieved documents, relying primarily on non-parametric knowledge. We also compare to “Closed-Book QA” approaches [52], which, like RAG, generate answers, but which do not exploit retrieval, instead relying purely on parametric knowledge. We consider four popular open-domain QA datasets: Natural Questions (NQ) [29], TriviaQA (TQA) [24], WebQuestions (WQ) [3] and CuratedTrec (CT) [2]. As CT and WQ are small, we follow DPR [26] by initializing CT and WQ models with our NQ RAG model. We use the same train/dev/test splits as prior work [31, 26] and report Exact Match (EM) scores. For TQA, to compare with T5 [52], we also evaluate on the TQA Wiki test set.

3.2 Abstractive Question Answering

RAG models can go beyond simple extractive QA and answer questions with free-form, abstractive text generation. To test RAG’s natural language generation (NLG) in a knowledge-intensive setting, we use the MSMARCO NLG task v2.1 [43]. The task consists of questions, ten gold passages retrieved from a search engine for each question, and a full sentence answer annotated from the retrieved passages. We do not use the supplied passages, only the questions and answers, to treat

MSMARCO as an open-domain abstractive QA task. MSMARCO has some questions that cannot be answered in a way that matches the reference answer without access to the gold passages, such as “What is the weather in Volcano, CA?” so performance will be lower without using gold passages. We also note that some MSMARCO questions cannot be answered using Wikipedia alone. Here, RAG can rely on parametric knowledge to generate reasonable responses.

3.3 Jeopardy Question Generation

To evaluate RAG’s generation abilities in a non-QA setting, we study open-domain question generation. Rather than use questions from standard open-domain QA tasks, which typically consist of short, simple questions, we propose the more demanding task of generating Jeopardy questions. Jeopardy is an unusual format that consists of trying to guess an entity from a fact about that entity. For example, “The World Cup” is the answer to the question “In 1986 Mexico scored as the first country to host this international sports competition twice.” As Jeopardy questions are precise, factual statements, generating Jeopardy questions conditioned on their answer entities constitutes a challenging knowledge-intensive generation task.

We use the splits from SearchQA [10], with 100K train, 14K dev, and 27K test examples. As this is a new task, we train a BART model for comparison. Following [67], we evaluate using the SQuAD-tuned Q-BLEU-1 metric [42]. Q-BLEU is a variant of BLEU with a higher weight for matching entities and has higher correlation with human judgment for question generation than standard metrics. We also perform two human evaluations, one to assess generation factuality, and one for specificity. We define factuality as whether a statement can be corroborated by trusted external sources, and specificity as high mutual dependence between the input and output [33]. We follow best practice and use pairwise comparative evaluation [34]. Evaluators are shown an answer and two generated questions, one from BART and one from RAG. They are then asked to pick one of four options—question A is better, question B is better, both are good, or neither is good.

3.4 Fact Verification

FEVER [56] requires classifying whether a natural language claim is supported or refuted by Wikipedia, or whether there is not enough information to decide. The task requires retrieving evidence from Wikipedia relating to the claim and then reasoning over this evidence to classify whether the claim is true, false, or unverifiable from Wikipedia alone. FEVER is a retrieval problem coupled with an challenging entailment reasoning task. It also provides an appropriate testbed for exploring the RAG models’ ability to handle classification rather than generation. We map FEVER class labels (supports, refutes, or not enough info) to single output tokens and directly train with claim-class pairs. Crucially, unlike most other approaches to FEVER, we do not use supervision on retrieved evidence. In many real-world applications, retrieval supervision signals aren’t available, and models that do not require such supervision will be applicable to a wider range of tasks. We explore two variants: the standard 3-way classification task (supports/refutes/not enough info) and the 2-way (supports/refutes) task studied in Thorne and Vlachos [57]. In both cases we report label accuracy.

4 Results

4.1 Open-domain Question Answering

Table 1 shows results for RAG along with state-of-the-art models. On all four open-domain QA tasks, RAG sets a new state of the art (only on the T5-comparable split for TQA). RAG combines the generation flexibility of the “closed-book” (parametric only) approaches and the performance of “open-book” retrieval-based approaches. Unlike REALM and T5+SSM, RAG enjoys strong results without expensive, specialized “salient span masking” pre-training [20]. It is worth noting that RAG’s retriever is initialized using DPR’s retriever, which uses retrieval supervision on Natural Questions and TriviaQA. RAG compares favourably to the DPR QA system, which uses a BERT-based “cross-encoder” to re-rank documents, along with an extractive reader. RAG demonstrates that neither a re-ranker nor extractive reader is necessary for state-of-the-art performance.

There are several advantages to generating answers even when it is possible to extract them. Documents with clues about the answer but do not contain the answer verbatim can still contribute towards a correct answer being generated, which is not possible with standard extractive approaches, leading